

UML 2.0 Component Diagram: Self-Service Coffee Kiosk System

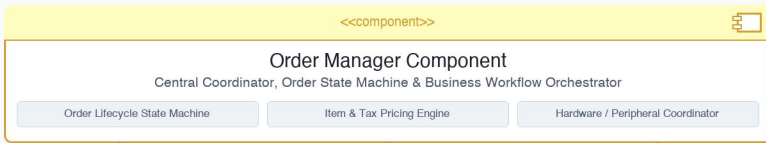
Architectural Pattern: Three-Tier Layered Architecture (Presentation, Business, Data Layers)

<<layer>> Presentation Layer



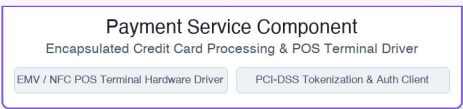
IOrderProcessing <<interface>>
API: submitOrder(items), cancelOrder(), getMenu()
Protocol: Direct in-Process API / Event Bus Call

<<layer>> Business Layer



IPaymentService <<interface>>
API: authorizePayment(orderId, amount)
Protocol: Synchronous Local IPC / POS Terminal Driver

<<component>>



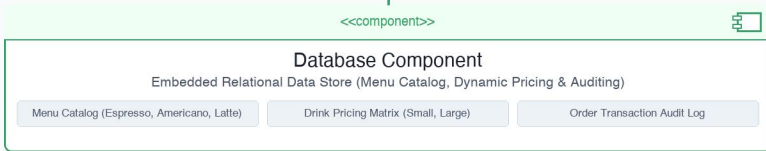
IReceiptPrinter <<interface>>
API: printReceipt(orderData), checkStatus()
Protocol: ESC/POS Command Stream over USB/Serial

<<component>>



IDataAccess <<interface>>
API: getMenuCatalog(), getItemPrice(), saveOrderRecord()
Protocol: Embedded SQL / SQLite API / DAO Queries

<<layer>> Data Layer



UML 2.0 Notation Legend

- <<component>> Executable System Module
- Provided Interface (Ball / Offers Service)
- Required Interface (Socket / Needs Service)
- Assembly Connector (Ball & Socket Joint)
- Architectural Layer Boundary

Architecture Style: Layered Architecture

Strict Dependency Flow: Presentation -> Business -> Data

Lab 3 Verification Checklist

- [x] 5 Components (UI, Order, Payment, Printer, DB)
- [x] 4 Key Interfaces (IOrderProcessing, IPaymentService, etc.)
- [x] Proper UML Ball-and-Socket Notation Employed
- [x] <<component>> Stereotype & UML 2.0 Icons Used
- [x] Protocol & Data Flow Interactions Labeled
- [x] Credit Card PCI-DSS Isolated to Business Layer
- [x] Hardware Drivers Encapsulated from UI Layer